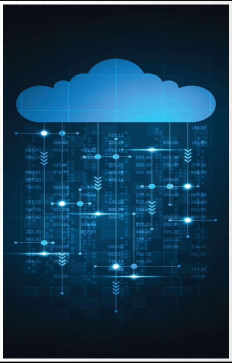


Cloud programming

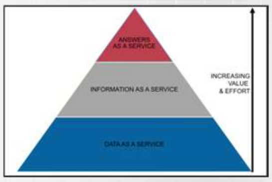
Rafał Palak

Wrocław University of Science and Technology

1

XaaS - anything as a service

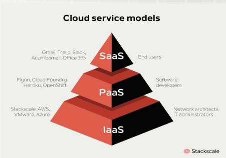
- Refers to any service that is available as a cloud service via the internet
- Includes SaaS, DaaS, PaaS, and IaaS etc.



2

Infrastructure as a Service (IaaS)

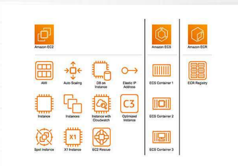
- A model in which virtual machines and servers are provided to customers for hosting a wide range of applications and IT services.
- Computing power, networking, and storage are delivered over the Internet.
- Examples: Amazon EC2, Rackspace, Google Compute Engine.

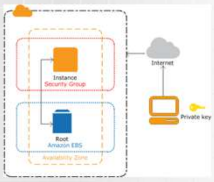


3

Amazon EC2 (Elastic Cloud Compute)

- A service that allows users to create a server in the AWS cloud.
- AWS refers to these servers as instances.
- Users can create, launch, and terminate server instances as needed.
- It provides users with control over the geographical location of the instances.

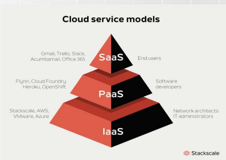




4

Platform as a Service (PaaS)

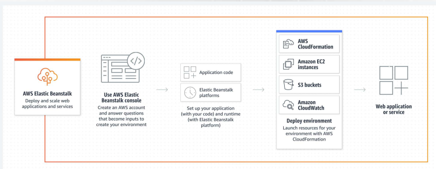
- A model that provides customers with a virtual platform for developing custom software.
- Tools are delivered over the Internet for building programs and applications.
- Examples: AWS Elastic Beanstalk, Microsoft Azure AppService, Google App Engine.



5

AWS Elastic Beanstalk

- AWS Elastic Beanstalk deploys web applications.
- It operates one layer of abstraction above the EC2 layer.



6

Elastic Beanstalk vs EC2

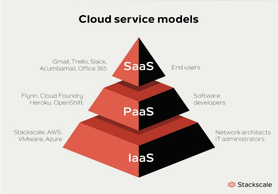
- Elastic Beanstalk sets up an “environment” for you, which can include multiple EC2 instances, an optional database, as well as several other AWS components such as an Elastic Load Balancer, Auto Scaling Group, and Security Group.
- It does not add any additional cost to the resources being used.

7

7

Software as a Service (SaaS)

- A model that delivers internet-based applications managed by a third party. Applications and programs are accessible and provided over the Internet. Examples: Dropbox, Slack, Spotify, YouTube, Microsoft Office 365, Gmail.

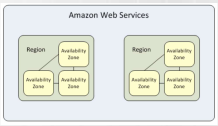


8

8

Region

- An area where data is stored.
- Each region is a separate geographic area.
- Every region is designed to be isolated from the others.
- When browsing resources (e.g., EC2 instances), only those associated with the selected region are visible.

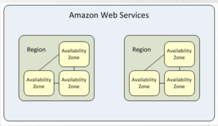


9

9

Availability Zone

- One or more data centers housing numerous servers.
- Each region contains multiple isolated locations called Availability Zones.
- Each Availability Zone is isolated, but zones within a region are connected through low-latency links.
- An Availability Zone is represented by the region code followed by a letter identifier, for example, us-east-1a.

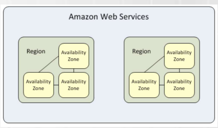


10

10

Edge locations

- AWS data centers designed to deliver services with the lowest possible latency.
- They are located closer to users than Regions or Availability Zones, often in major cities, enabling faster response times.
- Only certain latency-sensitive services use edge locations, such as CloudFront, Route 53, Web Application Firewall, and AWS Shield.



11

11

S3 (Amazon Simple Storage Service) [1]

- Implemented as an object that must be read from and written to by the application using it.
- Objects contain metadata — information about the object's attributes that helps the system catalog and identify it. Examples of objects include photos, videos, and music.
- Objects cannot be processed incrementally; they must be read and written in full — this can impact performance and consistency.

12

12

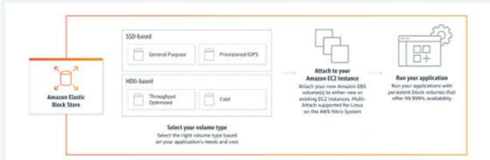
S3 (Amazon Simple Storage Service) [1]

- Provides storage through a web service interface.
- It can store any type of object, up to 5 TB each.
- Each object is stored in a bucket.
- Standardized REST and SOAP interfaces are supported.
- The default protocol is HTTP.

13

EBS (Amazon Elastic Block Store) [1]

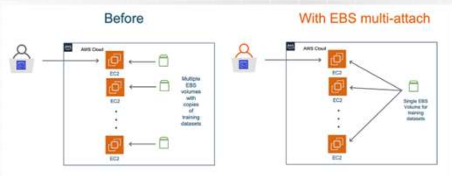
- "Hard drives for EC2" – Data storage for specific EC2 instances.



14

EBS (Amazon Elastic Block Store) [2]

- EBS volumes attached to instances are provided as storage volumes that persist independently of the instance's lifespan.
- They are recommended for data that needs to be quickly accessible and requires long-term durability.



15

EBS vs S3

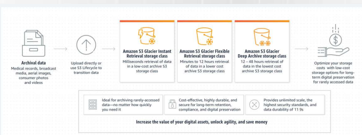
- EBS can only be used when it is attached to an EC2 instance.
- EBS cannot store as much data as S3.
- An EBS volume can be attached to only one EC2 instance at a time, and only that instance can access it.

- Access to S3 can be obtained independently.
- Data stored in S3 can be accessed by multiple EC2 instances.
- S3 experiences higher latency than Amazon EBS when writing data.

16

Amazon S3 Glacier

- Amazon S3 Glacier storage classes are designed for data archiving, offering the highest performance, greatest retrieval flexibility, and the lowest cost for cloud-based archive storage.
- It allows you to choose from three archive storage classes optimized for different access patterns and retention durations:
  - S3 Glacier Instant Retrieval
  - S3 Glacier Flexible Retrieval
  - S3 Glacier Deep Archive



17

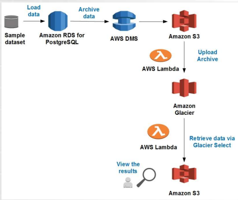
S3 Glacier Instant Retrieval

- Provides the lowest storage costs with millisecond retrieval times.
- Ideal for archival data that requires immediate access, such as medical images, genomic data, and news or reference data.

18

S3 Glacier Flexible Retrieval

- Provides retrieval options within a few minutes, or free bulk retrieval within 5–12 hours.
- Ideal for archival data that doesn't require immediate access but needs the flexibility for free retrieval of large datasets — such as backups or disaster recovery.

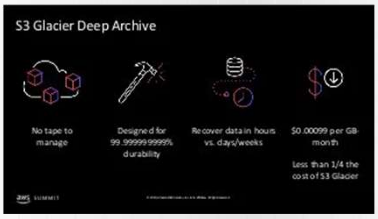


19

19

S3 Glacier Deep Archive

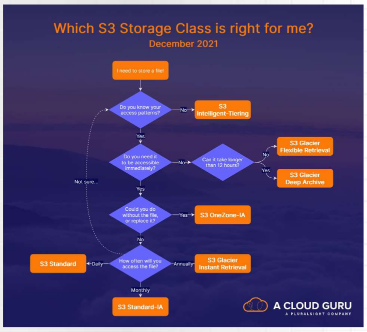
- Offers the lowest-cost cloud storage with data retrieval within twelve hours.
- Suitable for storing digital media archives.



20

20

What to choose?

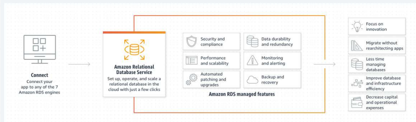


21

21

Amazon RDS (Amazon Relational Database Service)

- It is a collection of managed services that simplifies the configuration, operation, and scaling of cloud-based databases.
- This allows developers to create and manage relational databases in the cloud.
- Supported engines include Amazon Aurora (compatible with MySQL), Amazon Aurora (compatible with PostgreSQL), MySQL, MariaDB, PostgreSQL, Oracle, and SQL Server.



22

22

Amazon DynamoDB

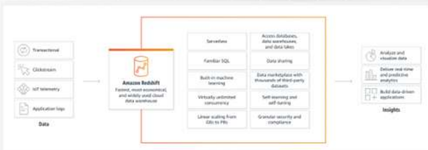
- AWS Non-Relational Database Service
- Data is stored in key-value pairs.
- It offers built-in security, continuous backups, automated multi-region replication, in-memory caching, and tools for data import and export.

23

23

Amazon Redshift

- AWS Data Warehouse Service
- This service can store massive amounts of data in a way that enables fast querying for business analytics purposes.
- It uses SQL to analyze structured and semi-structured data across data warehouses, operational databases, and data lakes.



24

24

RDS vs DB on EC2 - Administration

- Easy to set up.
- AWS automates the entire process of management, maintenance, and security.
- This allows you to focus on core tasks instead of routine upkeep.
- Multiple access options are available for relational database capabilities: through the AWS Management Console, the AWS RDS Command Line Interface (CLI), or simple REST API calls.

- Full control over the installed operating system, database version and configuration, as well as other software components.
- It requires handling all routine maintenance tasks, including patching, updates, backups, replication, and clustering.

25

25

RDS vs DB on EC2 - Availability

- It has built-in high availability.
- It automatically creates a primary database instance and replicates the data side-by-side to a standby instance in a different AWS Availability Zone.

- The user is responsible for configuring the database server in a high-availability cluster.

26

26

RDS vs DB on EC2 - Backups

- You can configure automatic backups.
- AWS CloudWatch can be used to receive events related to backup failures, completions, and more.
- On-demand database snapshots can be created and stored for as long as needed.

- Backups must be performed manually.
- CloudWatch cannot be used – separate monitoring is required to ensure that backups are performed regularly.

27

27

RDS vs DB on EC2 - Scalability

- Seamlessly integrates with Amazon scaling tools to support both vertical and horizontal scaling.
- Vertical scaling to a larger or more powerful instance can be done with just a few clicks.
- For horizontal scaling, launching additional read replicas can be automated, allowing the system to instantly respond to increasing demands for read-only workloads.

- Scalability must be configured manually.
- This process may involve setting up multiple EC2 instances, load balancing between them, configuring availability groups, sharding, and more.

28

28

RDS vs DB on EC2 - Security

- Offers encryption both at rest and in transit.
- Storage for database instances, read replicas, automated backups, and snapshots is encrypted while at rest.

- Encryption is performed at the EBS volume level, and database-level encryption can also be configured.

29

29

RDS vs DB on EC2 - Cost

- It is typically more expensive.
- AWS handles routine tasks.

- Installing and managing a database server on EC2 is usually more cost-effective than using RDS.
- However, it requires performing routine management tasks manually, such as backups, recovery, patching, and load management.

30

30

Wrocław University of Science and Technology

Thank you for your attention

31

31